

# YOUNGJI KOH

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🌐 [youngji-koh.github.io](https://youngji-koh.github.io)

## SUMMARY

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Researcher specializing in human–AI interaction, agentic and multi-agent systems, and multimodal computing for digital mental health and wellbeing.

## RESEARCH INTEREST

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Human–AI Interaction • Agentic and Multi-Agent Systems • Digital Mental Health • Multimodal Behavioral Modeling • LLM-based Contextual Reasoning • Ubiquitous Computing

## EDUCATION

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### Ph.D. in School of Computing

Sep. 2021 – Aug. 2026

KAIST, Daejeon, Republic of Korea

*Interactive Computing Laboratory (Advisor: Dr. Uichin Lee)*

**Dissertation:** Multimodal Behavior Modeling and AI Agent Systems for Mental Healthcare in Intelligent Environments.

### M.S. in School of Computing

Sep. 2019 – Aug. 2021

KAIST, Daejeon, Republic of Korea

*Software Architecture Laboratory (Advisor: Dr. Sungwon Kang)*

**Dissertation:** Deep Learning based Bug Report Summarization using Sentence Relations

### B.S. in Computer Science and Engineering

Mar. 2015 – Aug. 2019

Ewha Womans University, Seoul, Republic of Korea

GPA: 4.02/4.30 (*Summa Cum Laude*)

## RESEARCH EXPERIENCE

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### Visiting Researcher (Postdoctoral Fellow)

Sep. 2026 – Aug. 2027

*Interactive Intelligent Systems Laboratory, The University of Tokyo (Host: Dr. Koji Yatani)*

**Fellowship:** National Research Foundation of Korea (NRF) Postdoctoral Fellowship.

- Conducting research on human–AI interaction and AI-based mental healthcare systems.
- Investigating personalized and multi-agent AI approaches for supporting cognitive reframing, reflection, and mental wellbeing.

### Undergraduate Research Intern

May 2018 – May 2019

*Human-Computer Interaction Laboratory, Ewha Womans University (Advisor: Dr. Uran Oh)*

- Built a touchscreen-based mobile application providing object-level verbal descriptions to support independent artwork exploration by people with visual impairments.
- Designed a mobile application for supporting conflict management in romantic relationships on mobile messengers.

## FUNDED RESEARCH PROJECTS

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### Development of an Early Prediction System for Complex Diseases

Jun. 2025 – Aug. 2026

*Institute for ICT Planning & Evaluation (IITP), Lead Researcher*

- Designed a comorbidity-aware framework that coordinates disorder-specific agents for diagnosis and intervention planning.
- Focused on integrating LLM-based reasoning, multimodal sensing, and digital health applications for early intervention.

- **Keywords:** LLM Agent · Human-Centered Computing · Multimodal Sensing · Digital Health · Mental Health & Wellbeing

## AI Models for Multimodal Mental Health Prediction

Sep. 2023 – Aug. 2025

LG Electronics, Lead Researcher

- Developed deep learning and machine learning models using IoT, mobile, and wearable data for depression and anxiety prediction.
- Evaluated detection accuracy and reliability through large-scale, real-world deployments.
- **Keywords:** Machine Learning · Mental Health & Wellbeing · Multimodal Sensing · Digital Health

## IoT, Mobile, and Wearable Data Visualization for Mental Health Self-Reflection

May 2023 – Feb. 2025

LG Electronics, Lead Researcher

- Collected multimodal behavioral and physiological data via IoT, mobile, and wearable devices.
- Designed and evaluated personal informatics dashboards to promote self-reflection and wellbeing.
- The study was featured in national media outlets, including [Yonhap News](#), [Chosun Ilbo](#), [Herald Economy](#), and [Digital Times](#).
- **Keywords:** Data Visualization · IoT & Wearables · Mental Health · Ubiquitous Computing

## Robotic Counseling System for Mental Health Support

Feb. 2023 – Oct. 2024

LG Electronics, Research Contributor

- Supported a two-week field study with 20 participants evaluating emotional awareness, engagement, and sense of place in robotic counseling.
- **Keywords:** Social Robots · Mental Health · Human-Robot Interaction · Digital Interventions

## Context-Aware Multimodal Smart Speaker for Mental Health Self-Tracking

Sep. 2021 – Sep. 2023

LG Electronics, Lead Researcher

- Designed and developed a context-aware multimodal smart speaker for in-home mental health self-tracking.
- Led in-the-wild deployment for continuous behavioral monitoring and physiological data collection.
- The project received national media coverage highlighting its contribution to digital mental health innovation, including [Yonhap News](#), [Herald Economy](#), [Kookmin Ilbo](#), and [Maeil Business](#).
- **Keywords:** Smart Speaker · Context-Aware Sensing · Smart Home · Digital Health

## PUBLICATIONS

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### Preprints

#### [J2] A Multimodal Sensor Fusion Approach for Mental Health Detection Using Mobile, Wearable, and IoT Sensors

**Youngji Koh**, Gyuna Kim, Chanhee Lee, Panyu Zhang, Yunhee Ku, Inhwon Choi, Jewoo Ryu, and Uichin Lee  
Under review at *IEEE Journal of Biomedical and Health Informatics (J-BHI)*. IEEE. (Minor Revision Submitted)

### Journal & Conference

#### [C8] Exploring Data-Driven Approaches to Stress Management: A Systematic Review of Stress Tracking, Intervention, and System Evaluation Methods

**Youngji Koh**, Gwangyoung Lee, Jeonghyun Kim, Yugyeong Jung, Hwajung Hong, and Uichin Lee  
In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Barcelona, Spain, Apr. 13–17, 2026). CHI'26. ACM.

#### [C7] Leveraging Smartphone Human Interaction Routine Behavior Task Mining and Modeling for Daily Stress Monitoring

Hansoo Lee, Taehyeon Park, **Youngji Koh**, Jaegil Lee, and Uichin Lee  
In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*. UbiComp'26. ACM.

**[C6] Harnessing Home IoT for Self-tracking Emotional Wellbeing: Behavioral Patterns, Self-Reflection, and Privacy Concerns**

**Youngji Koh**, Chanhee Lee, Eunki Joung, Hyunsoo Lee, and Uichin Lee

In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* (Espoo, Finland, Oct. 14–16, 2025). UbiComp'25. ACM.

**[C5] 'In That Small Space with Just the Two of Us': User Experiences with Cumpa in a Robotic Counseling Center**

Chanhee Lee\*, Eunki Joung\*, **Youngji Koh**, Esther Kim, Sohwi Son, Sunjung Kwon, and Uichin Lee (\*: equal contribution)

In *Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)* (Bergen, Norway, Oct. 18–22, 2025). CSCW'25. ACM.

**[C4] Exploring Context-Aware Mental Health Self-Tracking Using Multimodal Smart Speakers in Home Environments**

Jieun Lim\*, **Youngji Koh**\*, Auk Kim, and Uichin Lee (\*: equal contribution)

In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Honolulu, Hawaii, USA, May 11–16, 2024). CHI'24. ACM.

**[C3] Interrupting for Microlearning: Understanding Perceptions and Interruptibility of Proactive Conversational Microlearning Services**

Minyeong Kim\*, Jiwook Lee\*, **Youngji Koh**, Chanhee Lee, Uichin Lee, and Auk Kim (\*: equal contribution)

In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Honolulu, Hawaii, USA, May 11–16, 2024). CHI'24. ACM.

**[J1] Deep Learning-based Bug Report Summarization Using Sentence Significance Factors**

**Youngji Koh**, Sungwon Kang, and Seonah Lee

*Applied Sciences*, 2022.

**[C2] Bug Report Summarization using Believability Score and Text Ranking**

**Youngji Koh**, Sungwon Kang, and Seonah Lee

International Conference on Artificial Intelligence in Information and Communication (ICAIIIC 2021). IEEE.

**[C1] Deep Learning Based Bug Report Summarization Using Sentence Assessment Score**

**Youngji Koh** and Sungwon Kang

Korea Software Congress (KSC 2020). *Best Paper Award*

**Poster & Workshop**

**[W2] Designing Clinician-Centered Explanations for Sensing-based Longitudinal Mental Health Monitoring**

Esther Kang, **Youngji Koh**, Seungwan Jin, and Uichin Lee

ACM UbiComp/ISWC 2026 XAI for U Workshop

**[W1] Data Visualization for Mental Health Monitoring in Smart Home Environment: A Case Study**

**Youngji Koh**, Chanhee Lee, Yunhee Ku, and Uichin Lee

IEEE VIS 2023 Workshop on Visual Analytics in Healthcare

**[P4] LV-Linker: Supporting Fine-grained User Interaction Analyses by Linking Smartphone Log and Recorded Video Data**

Hansoo Lee, Sangwook Lee, **Youngji Koh**, and Uichin Lee

ACM Symposium on User Interface Software and Technology (UIST 2022)

**[P3] Supporting Object-level Exploration of Artworks by Touch for People with Visual Impairments**

Nahyun Kwon, **Youngji Koh**, and Uran Oh

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2019)

**[P2] Analysis of Conflicts in Romantic Relationships on Mobile Messengers**

**Young-Ji Koh**, Hyunggu Jung, and Uran Oh

HCI Korea 2019

[P1] **Real-time Fire Evacuation System based on Indoor Location Tracking and Route Optimization**  
*Young-Ji Koh, Naeun Go, ChaeYoon Kim, SooHee Lee, and Sangsoo Park*  
Korea Software Congress (KSC 2018). *Student Paper Competition, Merit Award*

PATENTS

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*International Patent Applications*

[IP1] **Human Digital Twinning Method and System of Emotional Regulation for Emotional Labor in Workplaces Using Multi-Modal Sensor Data**  
*Uichin Lee, Duri Lee, Eunji Park, Youngji Koh, and Yunjo Hahn*  
U.S. Patent Application No. 18/212,435, filed Jun. 21, 2023; Publication No. US20240202623A1, published Jun. 20, 2024.

*Domestic Patents*

[DP3] **Apparatus for Collecting User's Experience According to the Situation and Method for Controlling Thereof**  
*Junhyung Kwon, Youngji Koh, Jieun Lim, and Uichin Lee*  
Korean Patent Application No. 10-2023-0060070, filed May 9, 2023; Publication No. 10-2024-0162897, published Nov. 18, 2024.

[DP2] **Human Digital Twinning Method and System of Emotional Regulation for Emotional Labor in Workplaces Using Multi-Modal Sensor Data**  
*Uichin Lee, Duri Lee, Eunji Park, Youngji Koh, and Yunjo Hahn*  
Korean Patent Application No. 10-2023-0038174, filed Mar. 23, 2023; Publication No. 10-2024-0092527, published Jun. 24, 2024.

[DP1] **System and Method for Providing Shortest Evacuation Route**  
*Sangsoo Park, Naeun Go, Youngji Koh, Chaeyoon Kim, and Soohee Lee*  
Korean Patent Application No. 10-2018-0162774, filed Dec. 17, 2018; Patent No. 10-2121672, granted Jun. 4, 2020.

HONORS AND AWARDS

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|----------------------------------------------------------------------|-------------|
| <b>Public Relations Award</b><br>Donga Science, Korea                | Dec. 2025   |
| <b>Creative Research Award</b><br>K-Data Science Conference, Korea   | Sep. 2025   |
| <b>Special Recognition for Outstanding Reviews</b><br>ACM CHI        | Dec. 2024   |
| <b>Ewha Womans University Scholarship</b><br>Merit-based scholarship | Spring 2018 |

TEACHING EXPERIENCE

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|----------------------------------------------------------------|------------------------|
| <b>Teaching Assistant, KAIST</b><br>CS492 Data Visualization   | Fall 2021, Spring 2023 |
| <b>Teaching Assistant, KAIST</b><br>CS204 Discrete Mathematics | Spring 2020            |

## COMMUNITY SERVICE

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**HCI@KAIST**

Student Steering Committee

2024, 2025

## ACADEMIC SERVICE

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### **Reviewer**

ACM Transactions on Computing for Healthcare

IEEE Transactions on Computational Social Systems

ACM UbiComp/ISWC Notes and Briefs

ACM CHI